Area Between Curves

ANSWER KEY

Section 1: Vertical Slices (Integration with Respect to x)

1 Find the area enclosed by $y = x^2$ and $y = 2x - x^2$.

a. 1/3 ✓

b. 2/3

c. 1/6

d. 1

Intersections: $x^2 = 2x - x^2 \Rightarrow 2x^2 - 2x = 0 \Rightarrow x = 0, 1$. Top curve is $2x - x^2$. Area $= \int_0^1 (2x - 2x^2)\,dx = [x^2 - \frac{2}{3}x^3]_0^1 = 1 - 2/3 = 1/3$.

2 Find the area between $y = x^2$ and $y = 4$.

a. 32/3 ✓

b. 16/3

c. 8

d. 64/3

Intersections at $x = \pm 2$. By symmetry: $2\int_0^2 (4 - x^2)\,dx = 2[4x - x^3/3]_0^2 = 2[8 - 8/3] = 2(16/3) = 32/3$.

3 Find the area between $y = \sqrt{x}$ and $y = x^2$.

a. 1/3 ✓

b. 2/3

c. 1/6

d. 1/2

Intersections at $x = 0$ and $x = 1$. $\int_0^1 (x^{1/2} - x^2)\,dx = [\frac{2}{3}x^{3/2} - \frac{x^3}{3}]_0^1 = 2/3 - 1/3 = 1/3$.

Section 2: Horizontal Slices (Integration with Respect to y)

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4 Find the area enclosed by $x = y^2$ and $x = y + 2$.

a. 9/2 ✓

b. 16/3

c. 7/2

d. 9

Intersections: $y^2 = y + 2 \Rightarrow y^2 - y - 2 = 0 \Rightarrow (y - 2)(y + 1) = 0 \Rightarrow y = -1, 2$. Area $= \int_{-1}^{2} (y + 2 - y^2)\, dy = [\frac{y^2}{2} + 2y - \frac{y^3}{3}]_{-1}^{2} = (2 + 4 - 8/3) - (1/2 - 2 + 1/3) = 10/3 - (-7/6) = 27/6 = 9/2$.

5 Why is integrating with respect to y advantageous for curves like $x = y^2 - 4$ and $x = 0$?

a. **The right curve minus left curve is a single continuous expression without piecewise splitting** ✓

b. Integration with respect to x is impossible

c. It changes the area value

d. The bounds are always 0 and 1

With horizontal slices, the right boundary is $x = 0$ and the left boundary is $x = y^2 - 4$ across the entire range $y \in [-2, 2]$, avoiding splitting into upper and lower halves.

6 Find the area enclosed between $x = 4 - y^2$ and the y-axis.

a. 32/3 ✓

b. 16/3

c. 8

d. 16

Intersections with y-axis ($x = 0$): $y = \pm 2$. Area $= \int_{-2}^{2} (4 - y^2)\, dy = 2[4y - y^3/3]_0^2 = 2[8 - 8/3] = 32/3$.

Section 3: Crossing Curves and Multiple Regions

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7 Find the total area enclosed between $y = \sin x$ and $y = \cos x$ from $x = 0$ to $x = \pi/2$.

a. $2\sqrt{2} - 2$ ✓

b. $\sqrt{2} - 1$

c. 2

d. 1

Intersection at $x = \pi/4$. On $[0, \pi/4]$, $\cos x > \sin x$; on $[\pi/4, \pi/2]$, $\sin x > \cos x$.
$\int_0^{\pi/4} (\cos x - \sin x)dx + \int_{\pi/4}^{\pi/2} (\sin x - \cos x)dx = (\sqrt{2} - 1) + (\sqrt{2} - 1) = 2\sqrt{2} - 2.$

8 Find the area enclosed between $y = x^3 - x$ and $y = 0$ for $x \in [-1, 1]$.

a. 1/2 ✓

b. 0

c. 1/4

d. 1

Due to odd symmetry, integrating directly gives 0 (signed area). Absolute area
$= 2\int_0^1 -(x^3 - x)\,dx = 2[\frac{x^2}{2} - \frac{x^4}{4}]_0^1 = 2(1/2 - 1/4) = 2(1/4) = 1/2.$

9 Find the area enclosed between $y = x^3$ and $y = 4x$.

a. 8 ✓

b. 4

c. 16

d. 0

Intersections: $x^3 = 4x \Rightarrow x(x-2)(x+2) = 0 \Rightarrow x = -2, 0, 2$. By symmetry, total area
$= 2\int_0^2 (4x - x^3)\,dx = 2[2x^2 - x^4/4]_0^2 = 2[8 - 4] = 2(4) = 8.$

Section 4: Applications and Boundaries

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10 Find the area of the region bounded by $y = e^x$, $y = e^{-x}$, and $x = 1$.

a. $e + 1/e - 2$ ✓

b. $e - 1/e$

c. $e + 1/e$

d. $e - 2$

On $[0, 1]$, $e^x \geq e^{-x}$. Area $= \int_0^1 (e^x - e^{-x})\,dx = [e^x + e^{-x}]_0^1 = (e + e^{-1}) - (1 + 1) = e + 1/e - 2$.

11 Find the area bounded by $y = \frac{1}{x}$, the x-axis, $x = 1$, and $x = e^2$.

a. 2 ✓

b. 1

c. e^2

d. $2e$

$\int_1^{e^2} \frac{1}{x}\,dx = [\ln x]_1^{e^2} = \ln(e^2) - \ln(1) = 2 - 0 = 2$.

12 The Gini index measures income inequality as twice the area between the line of perfect equality $y = x$ and the Lorenz curve $L(x) = x^2$ on $[0, 1]$. Find the Gini index.

a. $1/3$ ✓

b. $1/6$

c. $2/3$

d. $1/2$

Area $= \int_0^1 (x - x^2)\,dx = [x^2/2 - x^3/3]_0^1 = 1/2 - 1/3 = 1/6$. Gini index $= 2 \times 1/6 = 1/3$.

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